

Understanding Foundation Models

Instructor

Kunal Jain

Founder and CEO, Analytics Vidhya



Introduction to Foundation Models



Foundation Models

- Coined in 2001 by the Stanford Team
- Concept inspired from the self-supervised learning

Introduction to Self-Supervised Learning

Supervised Learning

- Performs well on labeled data
- Decent amount of labeled data is needed
- Label Creation is a manual process

Unsupervised Learning

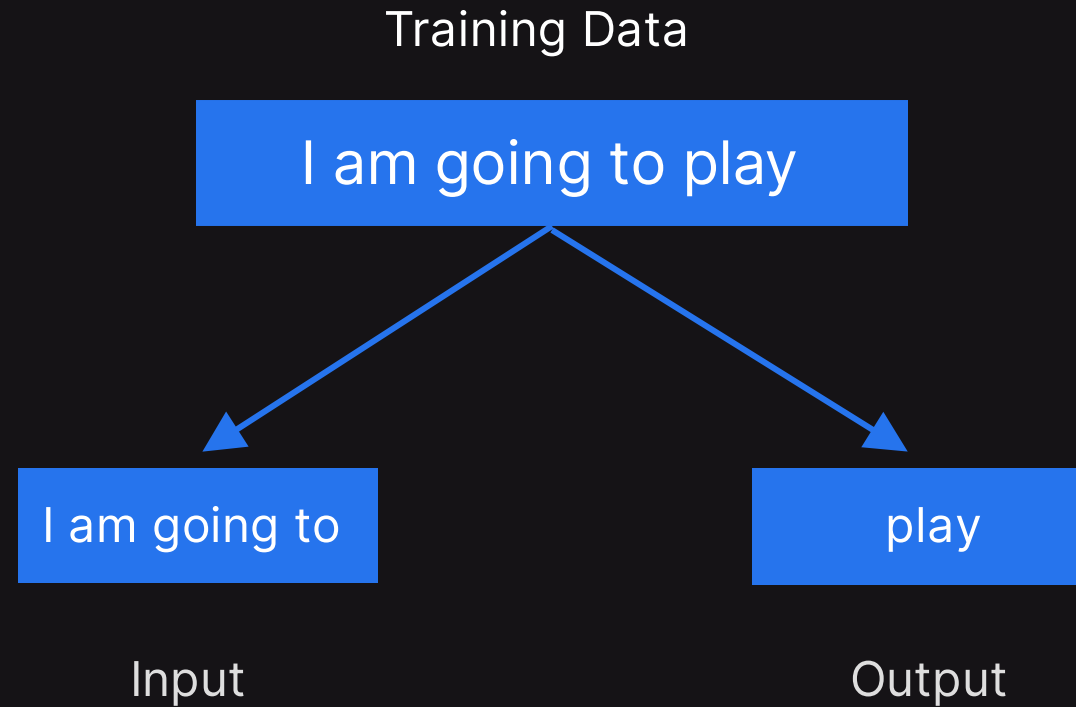
- Performs well on unlabeled data
- Not really effective compared to supervised learning

Self-Supervised Learning

- No external labeled dataset is needed
- Training labels are created for free from the unlabeled dataset

What is Self Supervised Learning?

Next word prediction



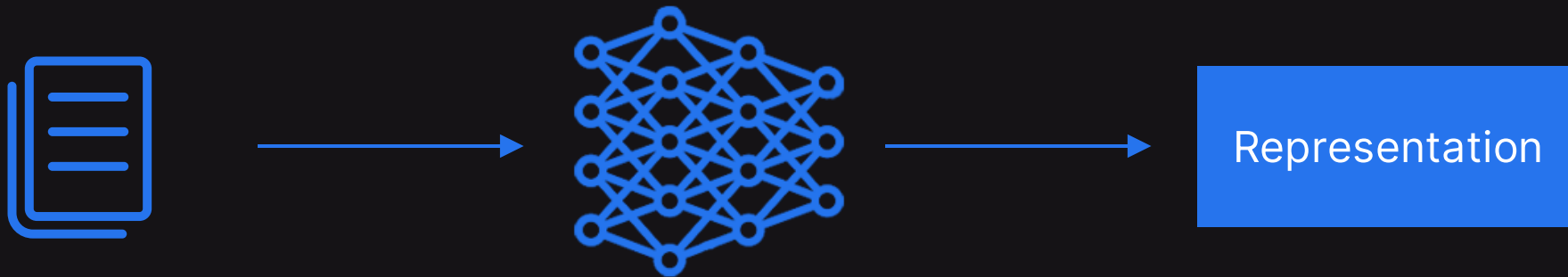
What is Self Supervised Learning?

Next word prediction



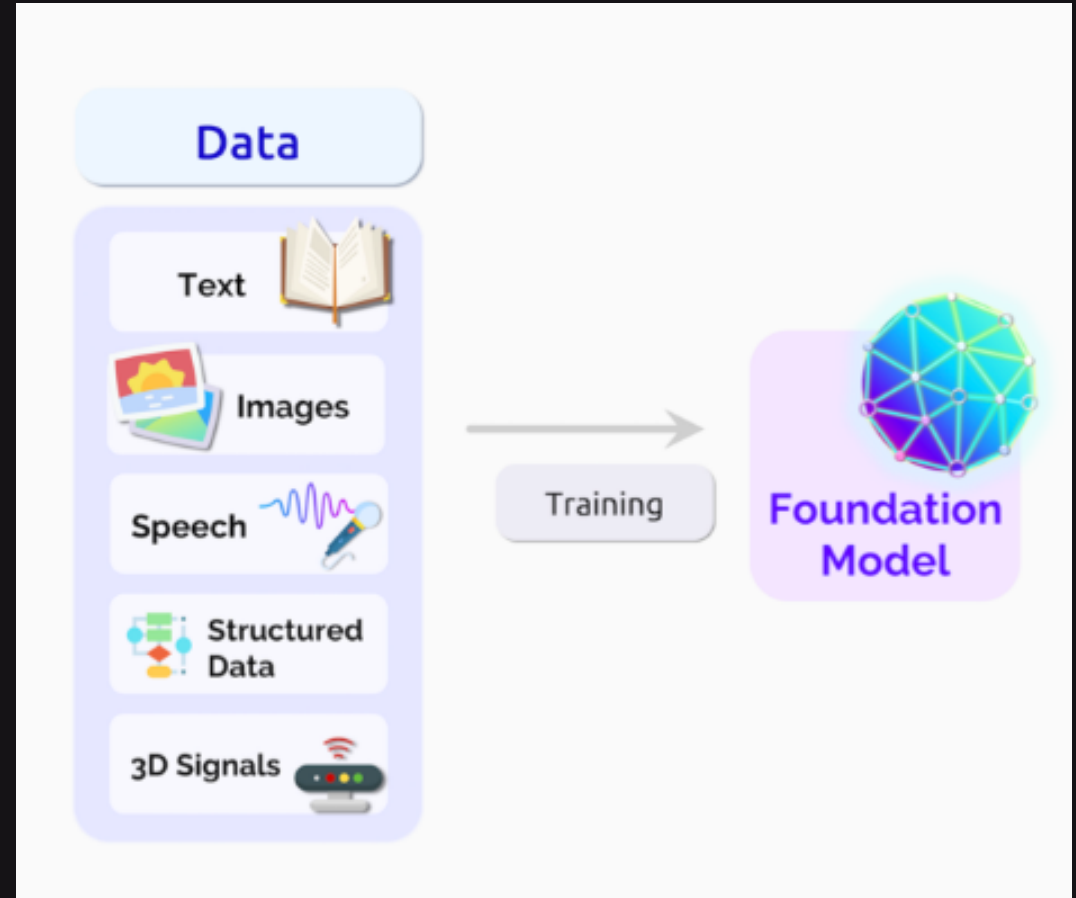
What is Self Supervised Learning?

Self Supervised Learning learns the representation form the unlabeled data.

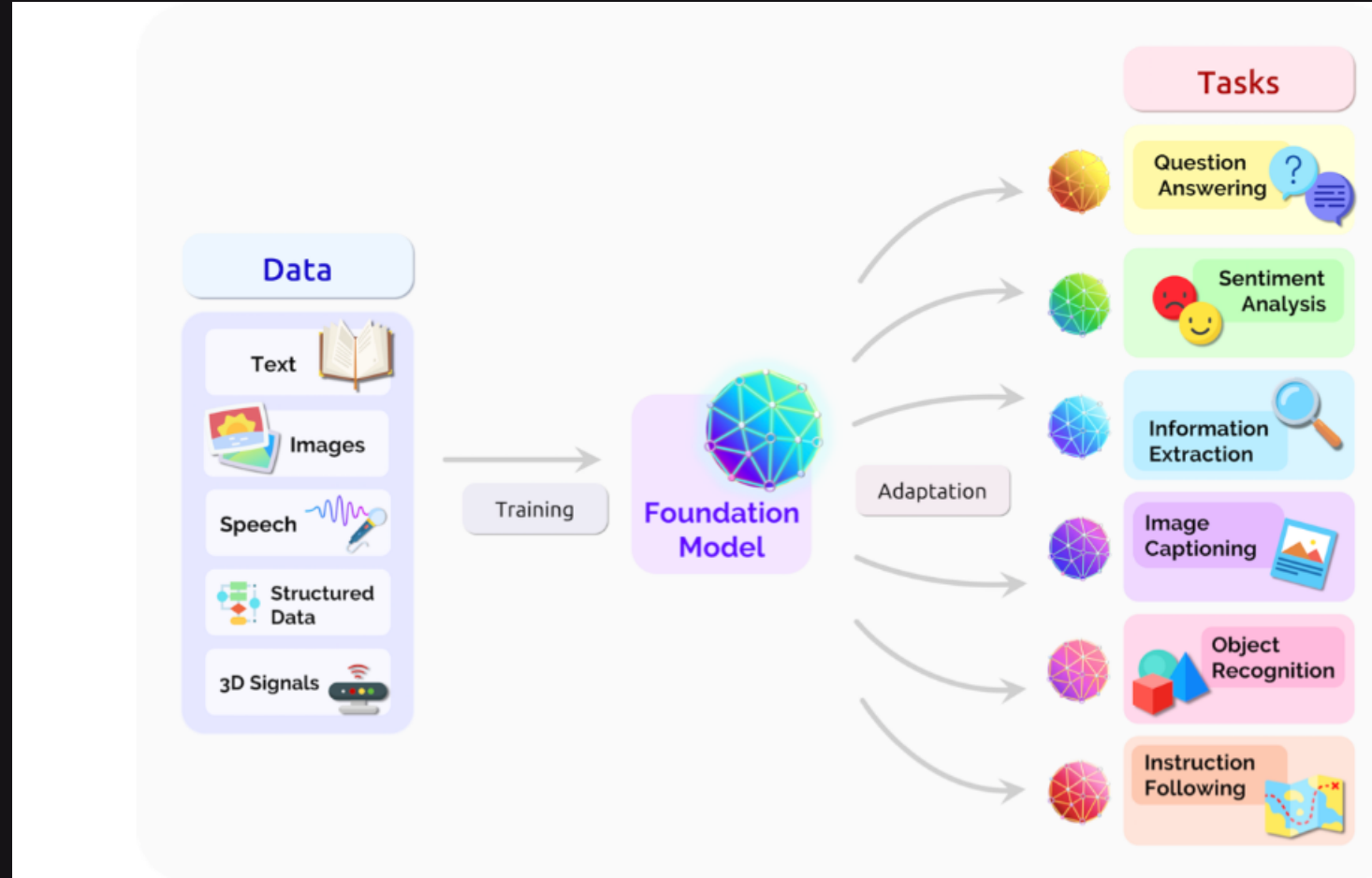


Understanding Foundation Model

AI model trains on very large "unlabeled" datasets using self-supervised learning.

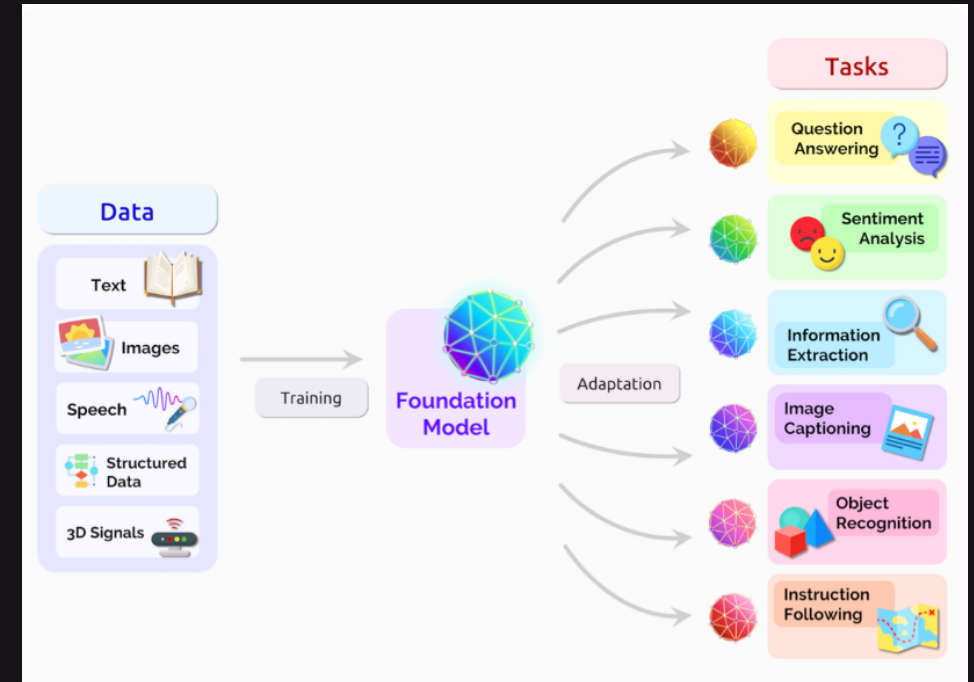


Understanding Foundation Model



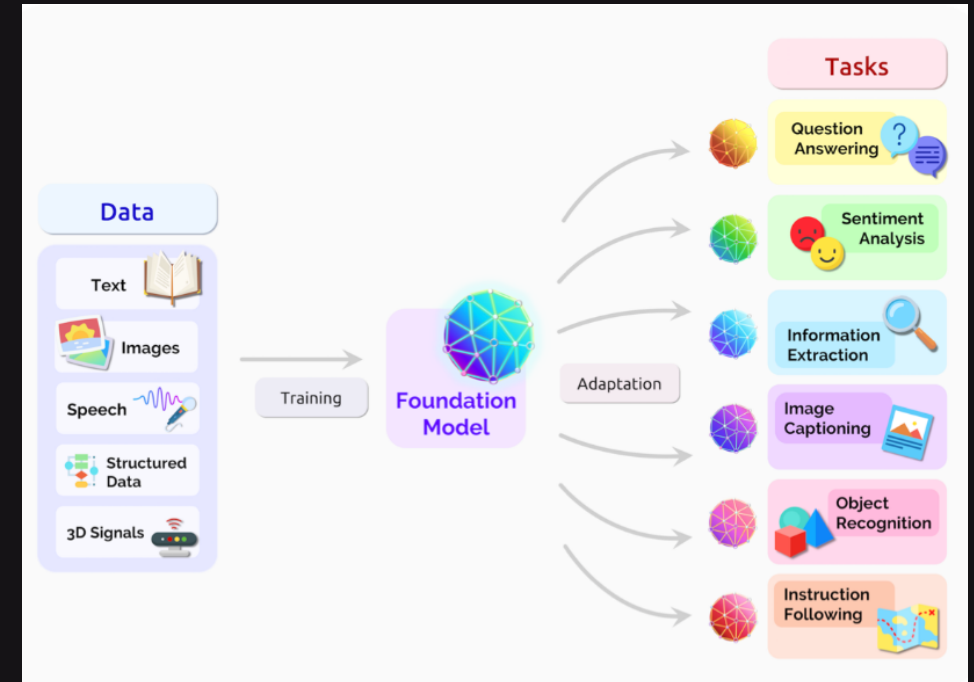
Why Foundation Model?

- Single model for various use cases



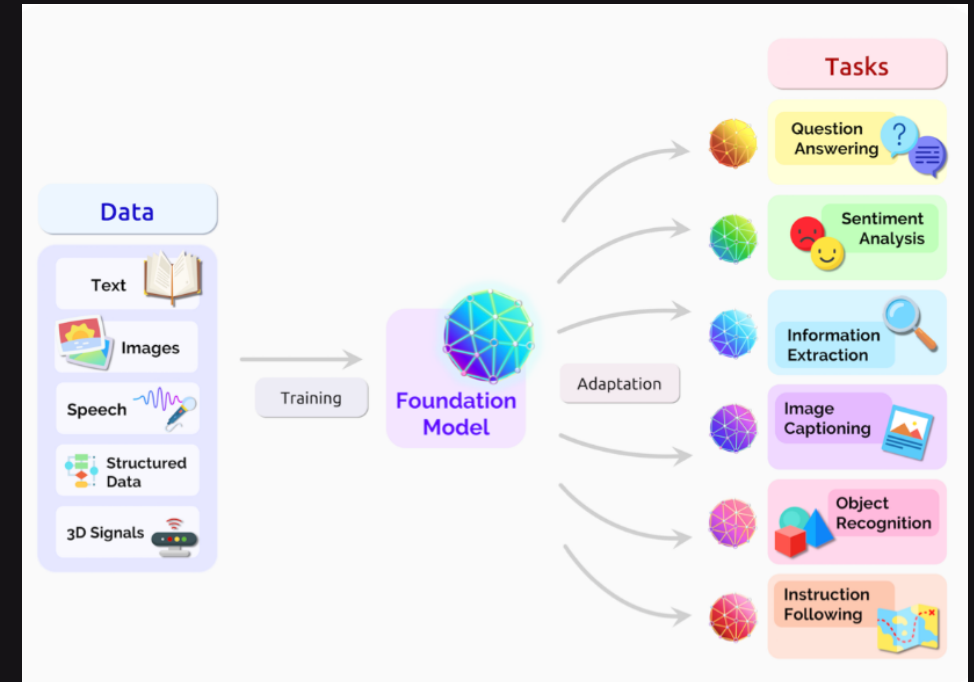
Why Foundation Model?

- Single model for various use cases
- High-performance model



Why Foundation Model?

- Single model for various use cases
- High-performance model
- Minimal task specific data



Example: Voice cloning Application

Before Foundation models

Minimum 10 hours of training data

After Foundation models

Just 3 minutes of training data

Different types of Foundation Models

Depending on the class of datasets, foundation models can be classified into 2 types:

Unimodal Foundation models

Multimodal Foundation models

Unimodal Foundation Models

Unimodal Foundation models

Deals with one class of dataset

Multimodal Foundation models

Examples of Unimodal Foundation Models



GPT-3



PaLM 2



Falcon LLM

Examples of Unimodal Foundation Models

Diffusion Models



Multimodal Foundation Models

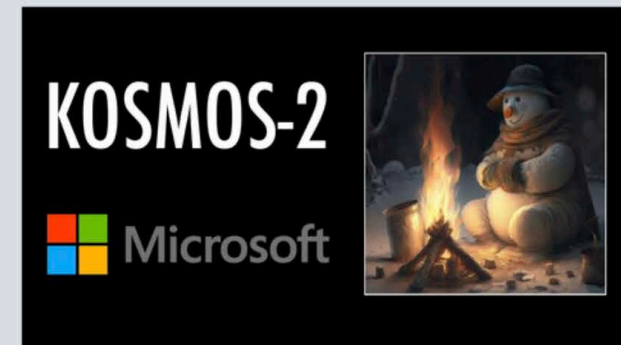
Unimodal Foundation models

Deals with one class of dataset

Multimodal Foundation models

Deals with more than one class of datasets

Examples of Multimodal Foundation Models



Thank You
